

VEHICLE PARKING SYSTEM

SRS ID	Requirement	Description	Priority
FR-01	User Registration/Login	The system shall allow users and parking administrators to register and log in securely.	High
FR-02	Vehicle Registration	The system shall allow users to add and manage vehicle details such as vehicle number, type, and owner information.	High
FR-03	Parking Slot Management	The system shall allow administrators to add, update, and remove parking slots.	High
FR-04	Slot Availability	The system shall display currently available and occupied parking slots.	High
FR-05	Vehicle Entry	The system shall record vehicle entry time and allocate an available parking slot.	High
FR-06	Vehicle Exit	The system shall record vehicle exit time and release the occupied parking slot.	High
FR-07	Parking Reservation	The system shall allow users to reserve an available parking slot for a specified period.	Medium
FR-08	Parking Fee Calculation	The system shall calculate parking fees based on parking duration and applicable rates.	High
FR-09	Payment Management	The system shall allow users to pay the calculated parking fee through supported payment methods.	High
FR-10	Parking History	The system shall maintain records of previous parking sessions, including entry time, exit time, slot, and fee.	Medium
FR-11	Notifications	The system shall notify users about reservation confirmation, parking expiry, and payment status.	Medium
FR-12	Admin Dashboard	The administrator shall be able to view parking occupancy, active vehicles, reservations, and revenue.	High

FR-13	Search Vehicle	The administrator shall be able to search for vehicles using vehicle number or other relevant details.	Medium
FR-14	Reports	The system shall generate reports related to parking usage, revenue, and vehicle activity.	Medium
NFR-01	Performance	The system should respond to user requests within an acceptable time under normal operating conditions.	High
NFR-02	Security	The system shall protect user credentials and prevent unauthorized access to system data.	High
NFR-03	Reliability	The system shall maintain accurate parking-slot and vehicle records without data loss.	High
NFR-04	Usability	The system shall provide a simple and intuitive interface for users and administrators.	Medium
NFR-05	Availability	The system should be available whenever parking management services are required.	High
NFR-06	Scalability	The system should support an increasing number of vehicles, users, and parking slots.	Medium
NFR-07	Maintainability	The system shall be designed so that parking rates, slot configurations, and other system settings can be modified easily.	Medium
NFR-08	Data Integrity	The system shall ensure that a parking slot cannot be assigned to multiple vehicles at the same time.	High